

| Project ID       | Domain                              | Ai Features                                                                                             |
|------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------|
| FSJ28-INTERN-035 | Digital Payments & FinTech Security | Isolation Forest Fraud Detection, Risk-Based Transaction Monitoring, Automated Security Account Locking |

Project Idea & Concept

WalletPay is an AI-powered Digital Payment and Secure Wallet System that connects users, contacts, support teams, and platform administrators. The system streamlines P2P payments through an interactive, conversation-based transfer interface (inspired by Google Pay and PhonePe). It features dynamic bank account linking for simulated deposits/withdrawals, a robust dispute resolution system, and an integrated real-time AI security engine that screens transactions for fraud to block unauthorized activity.

| Key Modules                                                                                                                                                  | Recommended Libraries & Tools                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| User & KYC Management<br>Interactive P2P Transfers<br>Bank Linking & Wallet Operations<br>Dispute & Support Desk<br>Admin Control Center<br>AI Fraud Monitor | <b>Backend:</b> Spring Boot, Spring Data JPA, Hibernate, Spring Security (Java 17+)<br><b>Frontend:</b> React, Vanilla CSS, Axios, Chart.js, React Router<br><b>Database:</b> MySQL<br><b>AI/ML Service:</b> Python, FastAPI, Uvicorn, Scikit-Learn (Isolation Forest), Joblib, Pandas<br><b>Visualization/Analytics:</b> Chart.js (Admin Timelines & KPI Stats)<br><b>Security:</b> BCrypt Hashing, UPI PIN Validation, Session Token Security |

AI / ML Feature Details

WalletPay integrates a Python-based Isolation Forest machine learning model that analyzes transaction features, including transaction amount, interaction frequency, and previous risk flags. When a user requests a transfer, the transaction is processed by the AI model. If anomalous behavior is detected, the AI flags the transaction as fraud, logs a security warning reason for the administrator, and automatically blocks the sender's account status to prevent further security breaches.

Recommended Libraries & Tools

Spring Boot, Spring Security, Hibernate, React, MySQL, FastAPI, and Scikit-Learn (Isolation Forest) are used to develop the core WalletPay application. Axios enables API communication, Chart.js provides analytics dashboards, while Maven and Git support dependency management, version control, and project development.

Implementation Flow

- Admin registers and manages the WalletPay platform.
- User registers using email and mobile OTP verification.
- User completes KYC verification by uploading Aadhaar and selfie for admin approval.
- User links simulated bank accounts and adds money to the wallet.
- User sends money through conversation-based P2P transfers using secure UPI PIN authentication.
- AI Fraud Monitor analyzes every transaction in real time and classifies it as **Normal**, **Suspicious**, or **Fraud**.
- Suspicious transactions are logged for investigation, while fraudulent transactions automatically trigger account protection.
- Users can raise disputes or submit unblock requests through the integrated support system.
- Admin reviews KYC requests, AI fraud alerts, support tickets, analytics dashboards, and securely manages user accounts.